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1 Probability stuff

1.1 The space of càdlàg functions (Skorokhod)

$\mathbb{D}([0,1])$ — *continue à droite, limite à gauche*

1.2 The Wiener measure and arcsine law

insert properties (and construction?) of the Wiener measure $W \in \mathcal{M}_1(\mathbb{D}([0,1]))$.

Let $\tau^+ \colon \mathbb{D}([0,1]) \to \mathbb{R}$ be the observable that returns the proportion of time its argument is positive, that is, $\tau^+(f) = \mathrm{Leb}\{x \in [0,1] : f(x) > 0\}$.

Lemma 1. *τ^+ is measurable and continuous in a full W-measure set.*

Proof. adapt this to $\mathbb{D}([0,1])$ This is proved in [1, appendix M15]. Since the normal distributions are non-atomic and for every $t \in [0,1]$, W_t has normal distribution, $W(\{f \in \mathbb{D}([0,1]) : f(t) = 0\}) = 0$ for all t, so by Fubini, $\mathrm{Leb}(\{t \in [0,1] : f(t) = 0\}) = 0$ for W-almost every f. It remains to check that τ^+ is continuous at such points $f \in W$. Indeed, 0 is the only discontinuity of the indicator function $1_{(0,\infty)}$, so if f is in this set and $f_n \to f$ uniformly, then $1_{(0,\infty)} \circ f_n \to 1_{(0,\infty)} \circ f$ almost everywhere, so by the DCT the same is true of their respective integrals. □

Theorem 1 (Arcsine law)**.** *For $\alpha \in [0,1]$,*

$$(\tau^+)_* W([0,\alpha]) = \frac{2}{\pi} \arcsin \sqrt{\alpha}.$$

1.3 Weak invariance principles

We will fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for the rest of this section. Given a sequence $\xi = (\xi_1, \xi_2, \dots)$ of random variables defined on Ω, we can define their partial sums $S_n^\xi = \sum_{i=1}^n \xi_i$ and a sequence of processes $D_n^\xi \colon \Omega \to \mathbb{D}([0,1])$ given by

$$D_n^\xi(\omega)(t) = \frac{S^\xi_{\lfloor nt \rfloor}(\omega)}{\sqrt{n}},$$

obtained by appropriately scaling the partial sums of the random variables up to some iterate n.

make proper figure

Figure 1: The process D_n^ξ

Remarkably, for many classes of sequences ξ, their corresponding processes D_n^ξ converge in some way to the Brownian motion — we denote by an *invariance principle* a result of this form. Perhaps the best known invariance principle is the Donsker's theorem, which is a *weak* invariance principle since it asserts convergence in the weak topology.

Theorem 2 (Donsker)**.** *If $\xi \subseteq L^2(\mathbb{P})$ is a sequence of i.i.d. variables with mean zero and variance 1, then the measures $(D_n^\xi)_*\mathbb{P}$ converge weakly to the Wiener measure. In other words, for every measurable continuous function $\varphi\colon \mathbb{D}([0,1]) \to \mathbb{R}$,*

$$\int \varphi \circ D_n^\xi \, d\mathbb{P} \to \int \varphi \, dW.$$

The motivation for looking into invariance principles is that they allow transferring certain results from Brownian motions to the sequence of partial sums S_n^ξ. In our case, the result we are interested in is the arcsine law, and since we know from lemma 1 that τ^+ is continuous a.e., weak convergence is enough:

Lemma 2. *Suppose $\xi \subseteq L^2(\mathbb{P})$ is such that $(D_n^\xi)_*\mathbb{P} \Rightarrow W$. Then the average of the time in which the random walk S_n^ξ is above $M \in \mathbb{R}$ converges to an arcsine distribution, in the sense that*

$$\lim_{n\to\infty} \mathbb{P}\left(\left\{\omega \in \Omega \;\middle|\; \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\} \le \alpha\right\}\right) = \frac{2}{\pi}\arcsin\sqrt{\alpha}.$$

Proof. Writing $\tau_n^+(\omega) = \frac{1}{n}\#\{1 \le i \le n \mid S_n^\xi(\omega) > M\}$, the left-hand side can be rewritten as $\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha])$. Set $c_n = M/\sqrt{n}$, and note that

$$\tau_n^+(\omega) = \frac{1}{n}\#\left\{1 \le i \le n \;\middle|\; \frac{S_i^\xi(\omega)}{\sqrt{n}} - \frac{M}{\sqrt{n}} > 0\right\} = \tau^+(D_n^\xi(\omega) - c_n).$$

Since by theorem 1 we know $Z = (\tau^+)_*W$ is equivalent to the Lebesgue measure on $[0,1]$, intervals are continuity sets for it. By the weak convergence assumption and continuity of τ^+, we know $(\tau^+ \circ D_n^\xi)_*\mathbb{P} \Rightarrow Z$, and since $c_n \to 0$ (as a sequence in $\mathbb{D}([0,1])$), in fact $(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P} \Rightarrow Z$. By Portmanteau, this means

$$\lim_{n\to\infty}(\tau_n^+)_*\mathbb{P}([0,\alpha]) = \lim_{n\to\infty}(\tau^+ \circ (D_n^\xi - c_n))_*\mathbb{P}([0,\alpha]) = \frac{2}{\pi}\arcsin\sqrt{\alpha}.$$

□

Lemma 3. *Under the same hypothesis as lemma 2, for all $M \in \mathbb{R}$ and for $\mathbb{P}$-almost every $\omega \in \Omega$,*

$$\lim_{n\to\infty} \frac{1}{n}\#\{1 \le i \le n \mid S_i^\xi(\omega) \in [-M, M]\} = 0.$$

Proof. Just like in the previous proof, write $\tau_n(\omega) = \frac{1}{n}\#\{1 \le i \le n : \mid S_n^\xi(\omega) \in [-M, M]\}$, and then note

$$\tau_n(\omega) = \tau^+(D_n^\xi(\omega) + c_n) - \tau^+(D_n^\xi(\omega) - c_n).$$

where $c_n = M/\sqrt{n}$. So

$$\lim \int \tau_n \, d\mathbb{P} = \int \tau^+ \, dW - \int \tau^+ \, dW = 0.$$

I haven't figured out how to continue from here, not even if the statement is true. □

1.3.1 Ergodic Martingale Differences

Donsker's theorem is far from being the most general result of its kind. In fact, it can be generalized to a weak invariance principle for ergodic martingale difference sequences, as stated below [see 1, theorem 18.3].

Definition 1. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables taking values in Y is said to be *stationary* if $\xi_*\mathbb{P}$ is invariant by the left shift $\sigma\colon Y^{\mathbb{N}} \to Y^{\mathbb{N}}$.

Definition 2. A sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables is said to be *ergodic* if $\xi_*\mathbb{P}$ is ergodic for the left shift.

Definition 3. Every sequence $\xi = (\xi_1, \xi_2 \dots)$ of random variables generates a filtration $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ of $\mathcal{F}$. We say that ξ is a *martingale difference sequence* if for all n,

$$\mathbb{E}_{\mathbb{P}}(\xi_n \mid \mathcal{F}_{n-1}) = 0.$$

There is a subtlety here, in the dynamical case the $\mathcal{F}_n$ gets smaller (in the sense $f^{-n}(\mathcal{F})$ is taken to be decreasing in other references). That is a reversed martingale difference, they seem to be more or less equivalent.

Theorem 3. *If $\xi \subseteq L^2(\mathbb{P})$ is a stationary ergodic martingale difference sequence for which $\sigma = \|\xi_1\|_2 > 0$, then the measures $(D_n^\xi)_*\mathbb{P} \Rightarrow \sigma W$.*

1.4 Martingale-coboundary decomposition

rewrite this section for ergodic processes in general, I think it will be less confusing later (see the remark in the next theorem)

The key idea for applying theorem 3 in a dynamical system $(X, \mathcal{F}, \mu, f)$, where μ is a f-invariant probability, is finding a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ for which $\varphi \in L^2(\mu)$ can be decomposed as a sum

$$\varphi = \varphi_0 + \phi \circ f - \phi \tag{1}$$

such that $\phi \in L^2(\mu)$, φ_0 is $\mathcal{G}$-measurable and $\eta = (\varphi_0 \circ f^{i-1})_{i\geq 1}$ is a reversed martingale difference sequence with respect to $(f^{-i}(\mathcal{G}))$. Indeed, this allows the transference of the weak invariant principle for the original φ: let $\xi = (\varphi \circ f^{i-1})_{i\geq 1}$, and notice that

$$S_n^\xi = S_n^\eta + \phi \circ f^n - \phi,$$

so

$$D_n^\xi(x)(t) = \frac{S^\xi_{\lfloor nt \rfloor}(x)}{\sqrt{n}} = D_n^\eta(x)(t) + \frac{1}{\sqrt{n}}\big(\phi \circ f^{\lfloor nt \rfloor} - \phi\big)(x).$$

It suffices to show that for μ-almost every x, the second term in the sum converges uniformly to zero (as a function of t). Given $\varepsilon > 0$, we look at the sets $B_n = \{x : (\phi^2 \circ f^n)(x) \geq n\varepsilon\}$. Since

$$\sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(\{\phi^2 \geq n\varepsilon\}) = \int \lfloor \phi^2/\varepsilon \rfloor \, d\mu \leq \frac{1}{\varepsilon}\|\phi\|_2^2,$$

by Borel-Cantelli, $\mu(\limsup_{n\to\infty} B_n) = 0$. If we take the intersection over k of $\liminf_{n\to\infty} B_n^c$ for $\varepsilon = 1/k$, we get a full-measure set G with the property that for every $x \in G$ and $\varepsilon > 0$, there exists N such that $n \geq N$ implies $\frac{1}{\sqrt{n}}|\phi(f^n(x))| \leq \varepsilon$. Now, let $M \geq N$ be big enough so that $n \geq M$ implies $\frac{1}{\sqrt{n}}|\phi(f^i(x))| \leq \varepsilon$ for all $0 \leq i \leq N$. So for $n \geq M$,

$$\sup_{t\in[0,1]} \frac{1}{\sqrt{n}}\big|\phi \circ f^{\lfloor nt \rfloor} - \phi\big|(x) \leq \max_{0\leq i\leq N} \frac{|\phi \circ f^i(x)|}{\sqrt{n}} + \max_{N\leq i\leq n} \frac{|\phi \circ f^i(x)|}{\sqrt{i}} \leq 2\varepsilon$$

as desired.

1.4.1 Existence of martingale-coboundary decomposition

From now on, we assume that μ is an ergodic measure for f.

Theorem 4. *Suppose $\varphi \in L^2(\mu)$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra such that $f^{-n}(\mathcal{G})$ is decreasing,*

$$\sum_{n=0}^{\infty} \|\mathbb{E}_\mu(\varphi \mid f^{-n}(\mathcal{G}))\|_2 < \infty \quad \text{and} \quad \sum_{n=0}^{\infty} \|\varphi - \mathbb{E}_\mu(\varphi \mid f^{n}(\mathcal{G}))\|_2 < \infty.$$

Then a decomposition as in eq. (1) exists. Moreover, this implies that

$$\int \varphi \, d\mu = \int \varphi_0 \, d\mu = 0$$

and that the series in

$$\sigma^2 = \|\varphi_0\|_2^2 = \|\varphi\|_2^2 + 2\sum_{n=1}^{\infty} \int \varphi \cdot \varphi \circ f^n \, d\mu$$

converges absolutely. By the preceding arguments, if $\varphi_0 \neq 0$ then $(D_n^\xi)_* \mu \Rightarrow \sigma W$.

Proof. I'm having trouble translating between the reversed martigale differences and normal martingale differences. I understand how one implies the other, but the reason is a bit abstract (lift everything to the shift in $\mathbb{R}^{\mathbb{N}}$ and then take the natural extension to $\mathbb{R}^{\mathbb{Z}}$) so I've found it quite confusing to reinterpret it back in the original space and write the proof in terms of dynamics again. I think I'll restate and write the proof of this theorem in terms of ergodic sequences of random variables (as written in the book) and explain the translation later, with the current statement as a corollary. Or figure it out. □

Remark 1. If φ was a coboundary $\varphi = \phi \circ f - \phi$ to begin with, then it is clear that $S_n^\xi = \phi \circ f^n - \phi$, so the iterates of the skew product $F(x, y) = (f(x), y + \varphi(x))$ satisfy

$$F^n(x_0, y_0) = (f^n(x_0), \phi(f^n(x_0)) - \phi(x_0) + y_0) \subseteq \mathrm{Graph}(\psi)$$

and are thus contained in the graph of ψ, where $\psi(x) = \phi(x) - \phi(x_0) + y_0$. In particular, F is not ergodic for $\mu \otimes \mathrm{Leb}$.

Corollary 1. *Suppose φ is such that there exists a summable sequence $c_n > 0$ satisfying*

$$\int (\psi \circ f^n)\varphi \, d\mu \leq \|\psi\|_2 c_n$$

for all $\psi \in L^2(\mathcal{G})$, where $\mathcal{G}$ is as in theorem 4. Then

$$\sum_{n=0}^{\infty} \|\mathbb{E}(\varphi \mid f^{-n}(\mathcal{G}))\|_2 < \infty.$$

Remark 2. The book [2] proves this condition is satisfied for smooth expanding maps, Hölder-continuous φ and $\mathcal{G} = \mathcal{F}$, so this includes the $f(x) = kx$ maps. The second condition is trivial when $\mathcal{G} = \mathcal{F}$.

Remark 3. More generally, [2] also shows the decomposition (1) exists for uniformly hyperbolic f and Hölder φ, also by showing that a sum like in theorem 4 converges due to decay of correlations, then uses this to show the central limit theorem (but I have not read the proof). In this case, $\mathcal{G}$ has something to do with the attractor and the second condition follows from Hölder somehow.

the examples in the Billingsley book are worth checking. They include both the kx maps (hmm... I guess B-M didn't see it?) and the Gauss map.

2 Historic behavior for skew products

Let (X, μ) be a Borel probability space, where X is compact and metrizable, and $F\colon X \times I \to X \times I$ a skew product of the form

$$F(x, y) = (g(x), f_x(y))$$

where $g\colon X \to X$ is ergodic and measure-preserving. Suppose there is a homeomorphism $\gamma\colon I \to \overline{\mathbb{R}}$ and a measurable function $p\colon X \to \mathbb{R}$ satisfying $t_x \circ \gamma = \gamma \circ f_x$ for all $x \in X$, where $t_x(y) = y + p(x)$ is the translation by $p(x)$.

$$\begin{array}{ccc} I & \xrightarrow{f_x} & I \\ \downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\gamma} \\ \overline{\mathbb{R}} & \xrightarrow{t_x} & \overline{\mathbb{R}} \end{array}$$

Note there is no loss of generality in assuming $\gamma(0) = -\infty$ and $\gamma(1) = +\infty$.

Theorem 5. *If the weak invariance principle is valid for the pair (g, p), in the sense $(D_n^\xi)_*\mu \Rightarrow \sigma W$ for $\sigma > 0$ and $\xi = (p \circ g^{i-1})_{i\geq 1}$, then F has historic behavior. In fact, for μ-almost every $x \in X$ and every $y \in (0, 1)$,*

$$\{e_n(x, y)\}'_{n\in\mathbb{N}} = [\delta_-, \delta_+] \tag{2}$$

where $\delta_- = \mu \otimes \delta_0$ and $\delta_+ = \mu \otimes \delta_1$ and the limit points are taken in the weak topology.*

Moreover,

$$(e_n^y)_*\mu \Rightarrow \delta_* Z \tag{3}$$

where Z is the arcsine measure on $[0, 1]$, $e_n^y(x) = e_n(x, y)$ are the empirical measures and $\delta(\lambda) = \lambda\delta_- + (1 - \lambda)\delta_+$.

Proof. Fix $\delta \in (0, 1)$, and define

$$\begin{aligned} T(x, y) &= \limsup_{n\to\infty} \frac{1}{n} \#\{0 \leq i < n \mid f_x^{(i)}(y) > \delta\} \\ &= \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} 1_{(M_y,\infty)}(S_i^\xi(x)) \end{aligned}$$

where $M_y = \gamma(\delta) - \gamma(y)$. Let E be the full μ-measure set given by lemma 3.

Claim 1: $T(x, y_1) = T(x, y_2)$ for all $x \in E$, $y_1, y_2 \in (0, 1)$.

Indeed, set $M = \max\{|M_{y_1}|, |M_{y_2}|\}$ and let $G = \{i : |S_i^\xi(x)| > M\}$. Since $1_{(M_{y_1},\infty)}(t) = 1_{(M_{y_2},\infty)}(t)$ whenever $|t| > M$, for all $i \in G$,

$$1_{(M_{y_1},\infty)}(S_i^\xi(x)) = 1_{(M_{y_2},\infty)}(S_i^\xi(x)).$$

So

$$\begin{aligned} |T(x, y_1) - T(x, y_2)| &\leq \limsup_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} |1_{(M_{y_1},\infty)}(S_i^\xi(x)) - 1_{(M_{y_2},\infty)}(S_i^\xi(x))| \\ &\leq \limsup_{n\to\infty} \frac{1}{n} \#\{0 \leq i < n \mid i \notin G\} = 0. \end{aligned}$$

This proves the claim. Since T is invariant by F, for any $x \in g^{-1}(E)$,

$$T(x,y) = T(g(x), f_x(y)) = T(g(x), y).$$

By ergodicity, there is a full measure set $F \subseteq X$ and $t \in [0,1]$ for which $T(x,y) = t$ for all $x \in F$, $y \in (0,1)$.

We can use the arcsine law to evaluate this constant. Given $\varepsilon > 0$, fix some $y \in (0,1)$ and define

$$E_n = \left\{ x \in X \;\middle|\; \frac{1}{n}\#\{0 \le i < n \mid S_i^{\xi}(x) > M_y\} > 1 - \varepsilon \right\}.$$

By the lemma 2, we know $\lim_n \mu(E_n) = Z([1-\varepsilon, 1]) > 0$, so

$$\mu(\limsup_{n\to\infty} E_n) \ge \limsup_{n\to\infty} \mu(E_n) > 0.$$

This means there exists $x_0 \in F \cap \limsup E_n$, and since

$$\limsup_{n\to\infty} E_n = \{x \in X \mid T(x,y) > 1 - \varepsilon\}$$

we have $t = T(x_0, y) > 1 - \varepsilon$. But $\varepsilon > 0$ is arbitrary, so $T(x,y) = 1$ for every $x \in F$ and $y \in (0,1)$.

So far, we started with $\delta \in (0,1)$ and showed that for

$$T_\delta(x,y) = \limsup_{n\to\infty} \frac{1}{n}\#\{0 \le i < n \mid f_x^{(i)}(y) > \delta\},$$

there is $F_\delta \subseteq X$ with $\mu(F_\delta) = 1$ such that for $x \in F_\delta$, $y \in (0,1)$, $T_\delta(x,y) = 1$. But note that for fixed x, y, the function $\delta \mapsto T_\delta(x,y)$ is monotone. With this observation, it's easy to see that if we define

$$F = \bigcap_{i=1}^{\infty} F_{1/i} \cap F_{1-1/i},$$

then $\mu(F) = 1$ and $T_\delta(x,y) = 1$ for all $x \in F$ and $y, \delta \in (0,1)$.

In particular, this implies that if $x \in F$ and $y \in (0,1)$, then for any $\delta \in (0,1)$, letting $U_n = \{0 \le i < n \mid f_x^{(i)}(y) > \delta\}$, the limit supremum of the y coordinate satisfies

$$\limsup_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} f_x^{(i)}(y) \ge \limsup_{n\to\infty} \frac{1}{n}\sum_{i\in U_n} \delta = \delta,$$

so the left-hand side is equal to 1. The proof we have so far works analogously for the time averages of the iterates in which $f_x^{(i)}(y) < \delta$, and by doing so we get

$$\liminf_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} f_x^{(i)}(y) = 0.$$

Note that those are time averages of the continuous bounded function $\pi_2 \colon X \times I \to \mathbb{R}$, so historic behavior is proved.

We now focus on eq. (2). By Birkhoff, we can find a full measure set $F_1 \subseteq X$ such that for $x \in F_1$,

$$\frac{1}{n}\sum_{i=0}^{n-1} \phi(F^i(x,0)) \to \int_X \varphi(z,0)\, d\mu(z), \quad \frac{1}{n}\sum_{i=0}^{n-1} \phi(F^i(x,1)) \to \int_X \varphi(z,1)\, d\mu(z).$$

Given x in the full measure set $F \cap F_1$ and $y \in (0,1)$, we construct an increasing sequence $(n_k^-)_k \subseteq \mathbb{N}$ inductively, letting $n_0^- = 1$ and $n_{k+1}^- > n_k^+$ be any integer satisfying

$$\frac{1}{n_k^-}\#\{0 \le i < n_k^- \mid f_x^{(i)}(y) < 1/k\} > 1 - 1/k.$$

Analogously, we can pick an increasing sequence $(n_k^+)_k \subseteq \mathbb{N}$ for which

$$\frac{1}{n_k^+}\#\{0 \le i < n_k^+ \mid f_x^{(i)}(y) > 1 - 1/k\} > 1 - 1/k.$$

Claim 2:

$$e_{n_k^-}(x,y) \Rightarrow \delta_- \quad \text{and} \quad e_{n_k^+}(x,y) \Rightarrow \delta_+.$$

We only show the first convergence, since the second is analogous.

Let $\phi \in C(X \times I)$ and $\varepsilon > 0$. Since ϕ is uniformly continuous, there is some k_1 for which

$$\sup\big\{|\phi(x,1) - \phi(x,s)| : x \in X, s \in [0, 1/k_1]\big\} < \varepsilon.$$

Let $D = \{i \mid f_x^{(i)}(y) < 1/k_1\}$, and let N be big enough so that for $k \ge N$,

$$\left| \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,0)) - \int_X \varphi(z,0)\, d\mu(z) \right| < \varepsilon$$

and

$$\frac{1}{n_k^-}\#\{0 \le i < n_k^- \mid i \notin D\} < \varepsilon/\|\phi\|_\infty,$$

so we can estimate

$$\begin{aligned}
&|e_{n_k^-}(x,y)(\phi) - \delta_-(\phi)| \\
&\qquad = \left| \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,y)) - \int_X \varphi(z,0)\, d\mu(z) \right| \\
&\qquad \le \left| \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,y)) - \frac{1}{n_k^-} \sum_{i=0}^{n_k^- - 1} \phi(F^i(x,0)) \right| + \varepsilon \\
&\qquad \le \frac{1}{n_k^-} \sum_{\substack{0 \le i < n_k^- \\ i \in D}} \big|\phi(F^i(x,y)) - \phi(F^i(x,0))\big| + \frac{2}{n_k^-} \sum_{\substack{0 \le i < n_k^- \\ i \notin D}} \|\phi\|_\infty + \varepsilon \\
&\qquad \le \frac{1}{n_k^-} \sum_{\substack{0 \le i < n_k^- \\ i \in D}} \big|\phi(F^i(x,y)) - \phi(F^i(x,0))\big| + 3\varepsilon \\
&\qquad \le \varepsilon + 3\varepsilon = 4\varepsilon.
\end{aligned}$$

This completes the proof of the claim.

□

2.1 do we need ergodicity

I think ergodicity is perhaps a red herring. For instance, look at $F\colon I \times \mathbb{R} \to I \times \mathbb{R}$ defined by

$$F(x, y) = (\{3x\}, y + 2x - 1).$$

Note that since $\{3x\} = \{x + 2x - 1\}$, if $(x, y) = F^n(x_0, y_0)$ then $\{x - x_0\} = \{y - y_0\}$. This means every orbits is inside a "spiral" curve like figure 2 shows. Nevertheless, by the

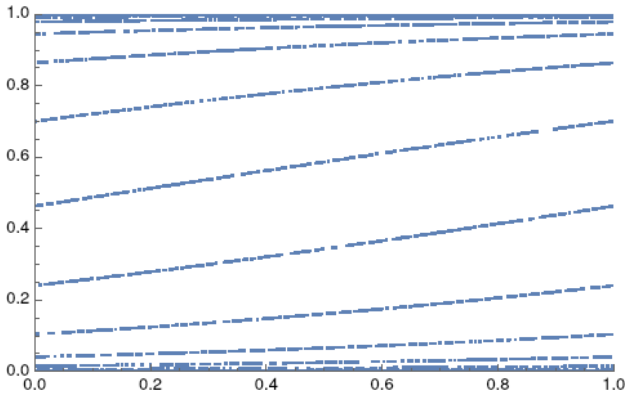

Figure 2: First 100 000 iterates of F, y axis is compactified.

theory we still have the weak invariant principle, and the y coordinates and their means plotted along time look like figure 3. The means don't seem to converge, and if this is the case then there is historic behavior.

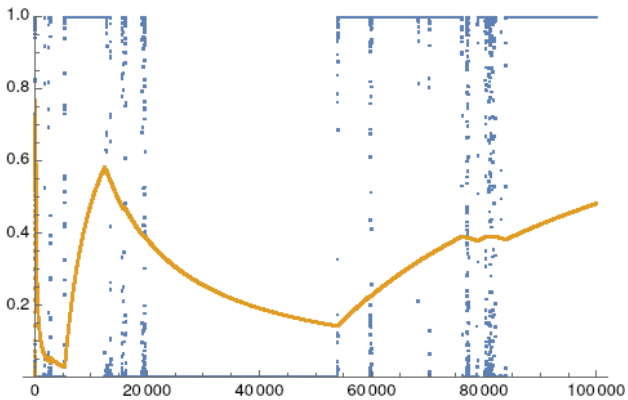

Figure 3: First 1 million iterates of the y coordinates (in blue) and their means (in orange).

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